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↳ [Btech CSE {Core}]

↳ DBMS Assignment

↳ Section- A:

Q.1→ Fill in the Blanks:

(i) The "Entity - Relationship" (ER). Diagram visually represents how Entities relate in a Database. offering a clear blueprint of the Data's Organizational Structure.

(ii) In the "Relational Model", data is Organized into Tables. with relationship between Entities defined Through Foreign Key Constraints.

(iii) The "Object- Oriented" Model treats data as a Collection of Objects, Each with Unique Attributes & methods. Mirroring real- world Entities and Interactions.

(iv) "Integrity" Constraints Ensures the reliability of data by enforcing rules that must be adhered to Maintain the Accuracy of Data.

↳ Section - B: [Case - Study Based]:

Q.2→ How Data Abstraction Simplifies Interaction between Healthcare professional & patient records.

Ans→ ↳ Data Abstraction in DBMS refers to hiding Complex database Structure & Showing Only relevant Information to Users. In Healthcare System, it plays an Important role in simplifying how doctors, Nurses & staff Interact with patient records.

↳ In a Hospital Environment, large Amount of patients data Such as Medical history, Diagnosis, Test results & billing Information are Stored in Database. If healthcare professionals had to deal with the Complex internal Structure of database, it would be difficult and Time-Consuming.

↳ Data Abstraction works at Three level:

i). Physical Level: A Doctor doesn't Need to know how data is Stored on the Server's hard drive or the specific Indexing Technique used.

ii). Logical Level: A Nurse sees a logical view of "Patient vitals", which Abstract away the backent table joins between "Patients" and "Medical_history".

iii). View Level: A Billing specialist Only Sees Insurance and payment Info, while a Surgeon sees Medical history and lab results, preventing Information Overload.

Q.3→ Benefits and Challenges of Implementing Data Abstraction in Healthcare Systems.

Ans→ ↳ Benefits:

i). Improved Data Security: Sensitive Information of patient is protected because user only Access Data relevant to their roles.

ii). Simplified Data Access: Doctors & Nurses Can Easily retrive patient Information without Understanding Complex Database Structures.

iii). Better Data Managment: Data Abstraction allows efficient organization & Managment of large Healthcare datasets.

↳ Challenges:

i). Implementation Complexity: Designing Multiple levels of abstraction requires Careful planning & Expertise.

ii). System Cost: Developing and Maintaining such systems Can be Expensive.

iii). Data Integration Issue: Integrating Data from different hospital Departments Can be challenging.

↳ Overall Despite these challenges. Data Abstraction significantly Improves the Efficiency and Security of Healthcare Database systems.

↳ Section → C [Simulation Based].

Q.3→ Design a University Database using ER Model.

Ans→ ↳ Entities:

i). Student Entity
- ↳ Student_ID [Primary Key]
- ↳ Student_Name
- ↳ Mobile_No.
- ↳ Department

ii). Course Entity
- ↳ Course_ID [Primary Key]
- ↳ Course_Name
- ↳ Credits

iii). Professor Entity
- ↳ Professor_ID [Primary Key]
- ↳ Professor_Name
- ↳ Email
- ↳ Specialization

↳ Relationships:

i). Enrollment Relationship [Student - Course]

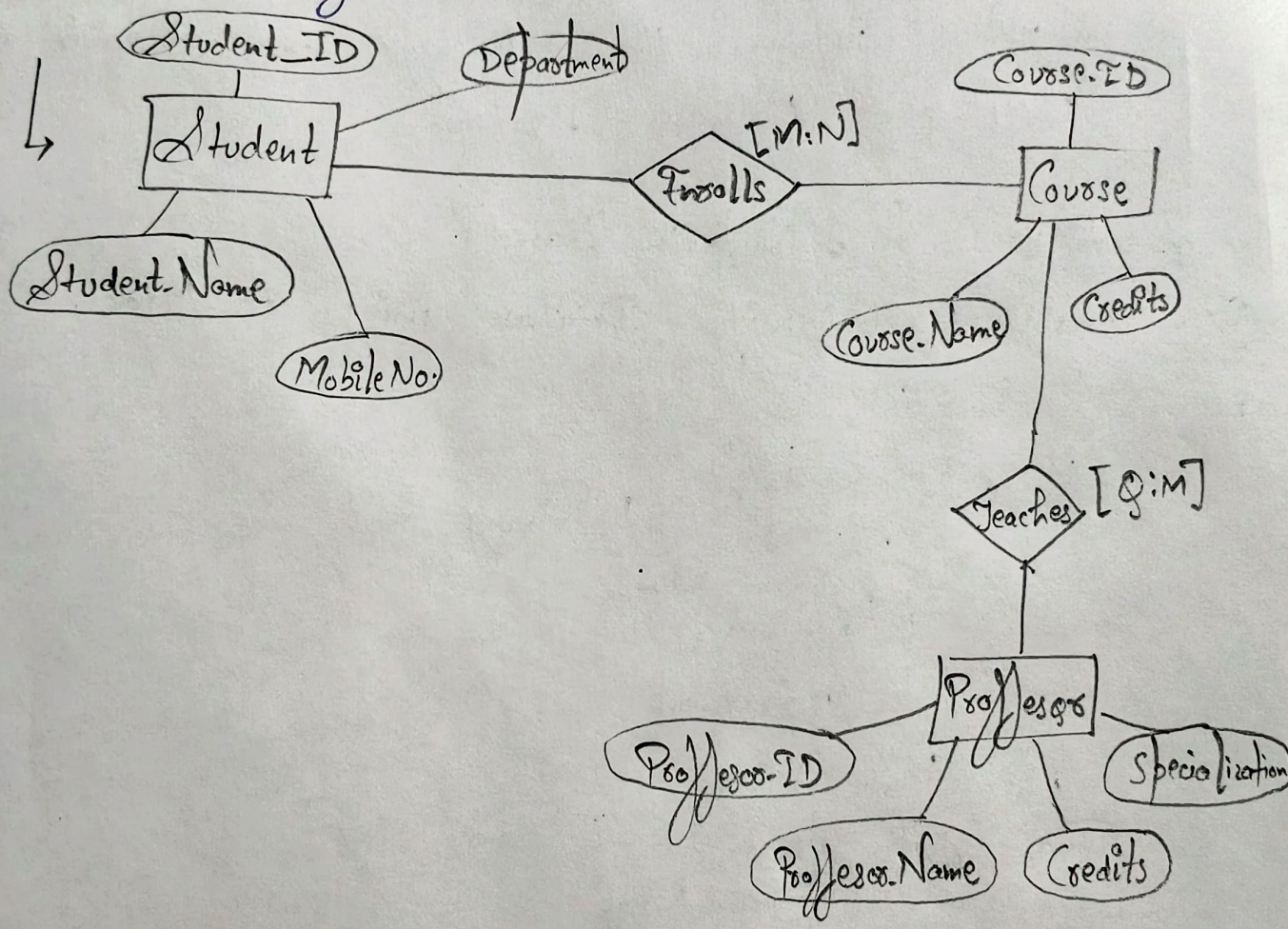
→ A student can enroll in multiple courses & a course can have many students. [Rel. Type → M:N].

ii). Teaching Relationship [Professor → Course]

↳ A professor can teach multiple courses but a course is taught by one professor.

→ [Rel. Type: One to Many].

↳ [E.R Diagram]:



↳ This ER Model helps organise University data efficiently, ensuring easy management of student enrollments, course details & professor information.